

Indian Statistical Institute

Mid semester examination (Part I)

Optimization and Numerical Methods
16 March 2025, 10:30am – 11:30am

Bachelor of Statistical Data Science
Maximum marks in Part I: 17

INSTRUCTIONS

Attempt as many questions as you can. You can score maximum 30 (combining part 1 and part 2).

Use of electronic devices is not allowed in the examination hall.

1. Let Z be a real-valued random variable with $\text{Prob}(Z = a_i) = p_i$, $i = 1, \dots, n$. Let $p \in \mathbb{R}^n$ lies in the standard probability simplex $P = \{p | 1^T p = 1, p \geq 0\}$.
 - (a) Using the definition of convexity, show that P is a convex set. [3]
 - (b) Verify if the following conditions yield convex sets of $p \in P$ or not. [2+2]
 - i. $E(Z^2) \leq d$,
 - ii. $\text{Var}(Z) \leq c$, where $\text{Var}(Z) = E(Z - E(Z))^2$ is the variance of Z .
2. Let $S \subseteq \mathbb{R}^n$ be the set of solutions for the quadratic inequality $S = \{x \in \mathbb{R}^n : x^T A x + b^T x \leq c\}$, where A is a symmetric positive definite matrix, $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$. Show that S is a convex set. [4]
3. Let $V = \{x \in \mathbb{R}^n : Ax = b\}$, where A is a $m \times n$ matrix and $b \in \mathbb{R}^m$ with $b \neq 0$. Show that V is an affine set, but **not** a convex cone. [3]
4. Check if $f(x) = \frac{1}{x^2}$ is a convex function on the domain $\mathbb{R} \setminus \{0\}$. [3]

Indian Statistical Institute

Mid semester examination (Part II)

Optimization and Numerical Methods
16 March 2025, 11:45am – 12:45pm

Bachelor of Statistical Data Science
Maximum marks in part II: 17

INSTRUCTIONS

Attempt as many questions as you can. You can score maximum 30 (combining part 1 and part 2).

Use of electronic devices is not allowed in the examination hall.

1. Check if the following functions are convex on the given domain:

(a) $f(x_1, x_2) = -\sqrt{x_1 x_2}$, $\text{dom}(f) = \mathbb{R}_{++}^2 = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, x_2 > 0\}$. [2.5]

(b) $f(x_1, x_2) = \frac{x_1 x_2}{x_1 + x_2}$, $\text{dom}(f) = \mathbb{R}_{++}^2 = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, x_2 > 0\}$. [2.5]

2. Prove the following statements

(a) If f is concave and positive, then $\frac{1}{f}$ is convex. [2.5]

(b) If f is convex, non-decreasing and positive valued function, and g is concave, non-increasing and positive, then $\frac{f}{g}$ is convex. [2.5]

3. Suppose $g: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable convex function with $\text{dom}(g) = \mathbb{R}$. Show that the running average $G: (0, \infty) \rightarrow \mathbb{R}$, defined as

$$G(x) = \frac{1}{x} \int_0^x g(t) dt,$$

is a convex function. [3]

4. (a) Show that the function $f(x) = x$ is log-concave on the domain $\text{dom}(f) = (0, \infty)$. Hence show that for any $a, b > 0$ and $0 \leq \theta \leq 1$, $a^\theta b^{(1-\theta)} \leq \theta a + (1-\theta)b$. [2]

(b) Using the inequality shown in (a), show that if $g: \mathbb{R}^n \rightarrow (0, \infty)$ is log-convex, then g is a convex function. [2]